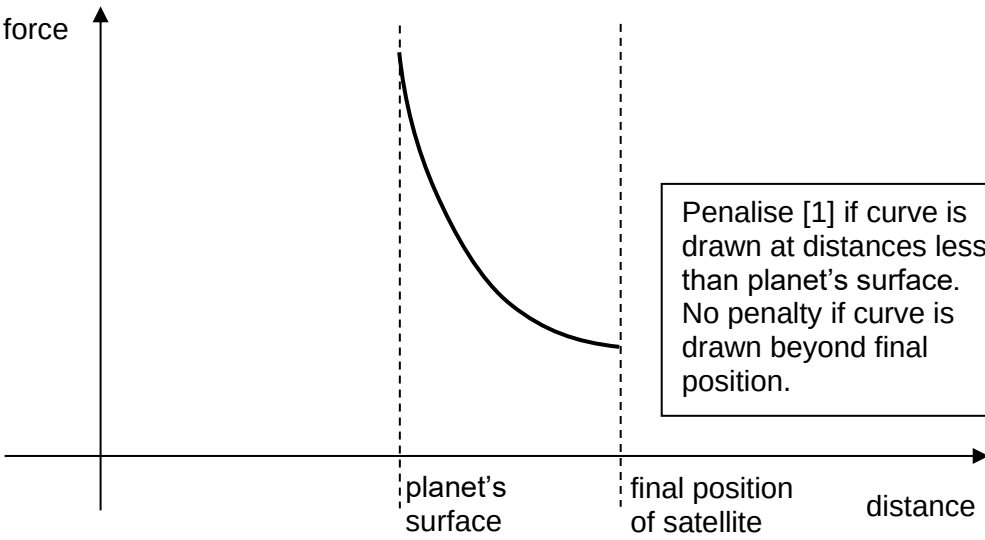


1a	similarity: lines are radial / greater separation of lines with increased distance from the sphere / lines are perpendicular to the surface of sphere difference: gravitational field lines directed towards sphere and electric field lines directed away from sphere	B1 B1
1bi	The gravitational potential at a point is the work done per unit mass in bringing a small test mass from infinity to that point.	B1 B1
1bii	(gravitational) potential is taken to be zero at infinity (gravitational) force is attractive work done by the external agent on the point mass moving it from infinity is negative	B1 B1 B1
1ci	$\text{GPE} = -\frac{GMm}{r}$ $\text{increase in GPE} = -\frac{GMm}{(6.40 + 0.900) \times 10^6} - \left(-\frac{GMm}{6.40 \times 10^6} \right)$ $= (6.67 \times 10^{-11})(6.00 \times 10^{24})(340) \left(\frac{1}{6.40 \times 10^6} - \frac{1}{(6.40 + 0.900) \times 10^6} \right)$ $= 2.62 \times 10^9 \text{ J}$	M1 A1
1cii	 • magnitude of force decreases non-linearly with distance • decreasing gradient, never infinite nor zero	B1 B1
1ciii	It represents the <u>increase</u> in potential energy of the satellite.	B1

2a	total volume of molecules is negligible compared with volume occupied by the gas	B1
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